PLSC 48110/28110: Lab and Field Experiments in Comparative Politics and Policy.

University of Chicago.

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Course description and objectives. Over the last several decades in what has become known as an "experimental revolution," social scientists have increasingly used experimental designs to study the effects of interventions and policies. The growing popularity of experimental methodology gives researchers leverage on causal questions, and bringing these designs to the field helps researchers and policy makers to take their research designs closer to the real social phenomena they seek to understand. In this course, we will study how these methods have been used to study politics and policy, with examples from Africa, Latin America, and India. We will ask what types of questions experiments can answer, and what types of things we can manipulate and measure. We will review design considerations, such as methods to account for treatment spillover, and randomization procedures. We will also read critiques of experimental methods and field experiments; students will discuss ethical considerations, and will consider tradeoffs to limitations of experimental designs in field settings. While we will carry out some calculations, this course will not be oriented towards statistical programming; however, undergraduate class participants should have taken PLSC 26969 or obtain instructor consent.

Assignments. Each student will be responsible for co-leading discussion on two of the second meetings of the week, beginning the third week of the course. For these meetings, discussion leaders will jointly provide a 20-30 minute total high-level overview of the papers

assigned, and will develop a list of discussion questions based on the course's running substantive and methodological themes. Co-discussion leaders may divide papers among themselves or may collaborate; presentation time is combined total across all papers and discussion leaders. Discussion questions should be submitted the evening before class.

The final project will be a pre-analysis plan for a hypothetical lab or field experiment (not survey experiment) relevant to comparative politics. In the sixth week of the class, students should submit one page overviews of the proposed research design, and will meet with the instructor to discuss.

Grade composition

- Leading class discussions: 25% for each of two classes
- Final project: 40%
- Participation: 10%

Accommodations. Please reach out to me directly if you would like to request accommodations for the course to better facilitate your learning. Student Disability Services (https://disabilities.uchicago.edu/) is also available to provide you resources and support, and may provide approval for specific academic accommodations. If you or your household are affected by the ongoing pandemic in a way that affects your ability to participate in or attend class, please reach out to me as well. Informing me in a timely manner will help me to ensure accommodations are met and I am able to implement an appropriate assessment of your learning.

Academic honesty. All students are expected to be familiar with and follow the University's academic honesty policies (https://studentmanual.uchicago.edu/academic-policies/academic-honesty-plagiarism/). For this course, you may use generative artificial intelligence tools such as ChatGPT for brainstorming, but you may not directly use any text generated from such tools in any assignments. Note that artificial intelligence may "hallucinate" information, so exercise extra caution and verify any information AI provides with reliable sources and proper citation practices (see discussions on AI hallucination in: https://www.nytimes.com/2023/03/29/technology/ai-chatbots-hallucinations.html, https://www.nytimes.com/2023/11/06/technology/chatbots-hallucination-rates.html).

Course schedule

Week 1: Course introduction

Class 1.1

Class overview. Social science questions we ask and answer with experiments. Randomization inference.

Readings

- None.
- For reference (you don't have to read these)
  - Holland, Paul W. (1986). Statistics and causal inference. *Journal of the American Statistical Association, 81*(396), 945-960. https://doi.org/10.1080/01621459.1986.10478354
  - King, Keohane, and Verba (1994). Designing Social Inquiry: *Scientific Inference in Qualitative Research.* Princeton University Press. Chapter 1. https://doi.org/10.1515/9781400821211-002

Week 2: Introduction to estimation, and survey experiments

Class 2.1

Overview of the recent history of field experiments for policy and politics, and some prospective considerations.

Readings

- Levitt, S. D., & List, J. A. (2008). Field experiments in economics: The past, the present, and the future. *European Economic Review*, 53(1), 1-18. https://doi.org/10.1016/j.euroecorev.2008.12.001

- Deaton, A. (2020). Randomization in the tropics revisited: a theme and eleven variations (No. w27600). National Bureau of Economic Research. https://doi.org/10.3386/w27600
- Banerjee, A., Duflo, E., Glennerster, R., & Kinnan, C. (2015). The miracle of microfinance? Evidence from a randomized evaluation. *American economic journal: Applied economics*, 7(1), 22-53. http://dx.doi.org/10.1257/app.20130533

Class 2.2

Estimation and inference for randomized experiments.

Readings

- Gerber, A. S., & Green, D. P. (2012). *Field experiments: Design, analysis, and interpretation*. WW Norton. Chapter 2: Causal Inference and Experimentation.
- Athey, S., & Imbens, G. W. (2017). The econometrics of randomized experiments. In *Handbook of economic field experiments* (Vol. 1, pp. 73-140). North-Holland. https://doi.org/10.1016/bs.hefe.2016.10.003. Parts 1-4 only.

Week 3: Lab and lab-in-the-field experiments

Class 3.2

Applications of behavioral games and lab-in-the-field experiments.

Readings

- Henrich, J., Boyd, R., Bowles, S., Camerer, C., Fehr, E., Gintis, H., & McElreath, R. (2001). In search of homo economicus: behavioral experiments in 15 small-scale societies. *American Economic Review*, 91(2), 73-78. https://doi.org/10.1257/aer.91.2.73
- Habyarimana, J., Humphreys, M., Posner, D. N., & Weinstein, J. M. (2007). Why does ethnic diversity undermine public goods provision?. *American political science review*, 101(4), 709-725. https://doi.org/10.1017/S0003055407070499

- Baldassarri, D., & Grossman, G. (2011). Centralized sanctioning and legitimate authority promote cooperation in humans. *Proceedings of the National Academy of Sciences*, 108(27), 11023-11027. https://doi.org/10.1073/pnas.1105456108

Week 4: Estimation considerations

Class 4.1

Experiments and covariate adjustment.

Readings

- Athey, S., & Imbens, G. W. (2017). The econometrics of randomized experiments. In *Handbook of economic field experiments* (Vol. 1, pp. 73-140). North-Holland. https://doi.org/10.1016/bs.hefe.2016.10.003. Parts 5-6 only.
- Lin, W. (2012). Regression Adjustment in Randomized Experiments: Is the Cure Really Worse Than the Disease?. Blog post.
  - Part I
  - Part II

Class 4.2

Measurement and results presentation.

Readings

- Olken, B. A. (2010). Direct democracy and local public goods: Evidence from a field experiment in Indonesia. *American political science review*, 104(2), 243-267. https://doi.org/10.1017/S0003055410000079
- Wantchekon, L. (2003). Clientelism and voting behavior: Evidence from a field experiment in Benin. *World politics*, 55(3), 399-422. https://doi.org/10.1353/wp.2003.0018

- Anderson, S. E., Buntaine, M. T., Liu, M., & Zhang, B. (2019). Non-governmental monitoring of local governments increases compliance with central mandates: a national-scale field experiment in China. *American Journal of Political Science*, *63*(3), 626-643. https://doi.org/10.1111/ajps.12428

Week 5: Design documents and pre-analysis plans

Class 5.1

Writing a design document or pre-analysis plan.

Readings

- Gerber, A. S., & Green, D. P. (2012). *Field experiments: Design, analysis, and interpretation*. WW Norton. Chapter 13: Writing an Experimental Proposal, Research Report, and Journal Article.
- United States. National Commission for the Protection of Human Subjects of Biomedical, & Behavioral Research. (1978). *The Belmont report: ethical principles and guidelines for the protection of human subjects of research*. Department of Health, Education, and Welfare, National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/read-the-belmont-report/index.html
- Design documents (for reference, you can skim these):
  - Young, L. E. (2015). The psychology of political risk: Replication and extension. https://doi.org/10.17605/OSF.IO/WGJZ3
  - Gaikwad, N., & Nellis, G. (2019). Field Experiment on the Political Empowerment of Internal Migrants in India. https://doi.org/10.17605/OSF.IO/7VTQH Also read supplementary information sections K and L, available at https://doi.org/10.7910/DVN/G1JCKK

Class 5.2

Applications from design documents, ethical considerations with human subjects research.

Readings

- Young, L. E. (2019). The psychology of state repression: Fear and dissent decisions in Zimbabwe. *American Political Science Review*, 113(1), 140-155. https://doi.org/10.1017/S000305541800076X
- Gaikwad, N., & Nellis, G. (2021). Overcoming the political exclusion of migrants: Theory and experimental evidence from India. *American Political Science Review*, 115(4), 1129-1146. https://doi.org/10.1017/S0003055421000435

Week 6: Design considerations: power and randomization

Class 6.1

Power calculations and randomization procedures.

Readings

- Athey, S., & Imbens, G. W. (2017). The econometrics of randomized experiments. In *Handbook of economic field experiments* (Vol. 1, pp. 73-140). North-Holland. https://doi.org/10.1016/bs.hefe.2016.10.003. Part 7 only.
- JPAL *Power calculations*. https://www.povertyactionlab.org/resource/power-calculations

Class 6.2

Different randomization procedures.

Readings

- Blair, R. A., Moscoso-Rojas, M., Castillo, A. V., & Weintraub, M. (2022). Preventing Rebel Resurgence after Civil War: A Field Experiment in Security and Justice Provision in Rural Colombia. *American Political Science Review*, 116(4), 1258-1277. https://doi.org/10.1017/S0003055422000284
- Blattman, C., Duncan, G., Lessing, B., & Tobon, S. (2022). State-building on the margin: An urban experiment in Medellín (No. w29692). *National Bureau of Economic Research*. https://doi.org/10.31235/osf.io/3bncz

Assignment: Write up one-page research proposal. Meet with instructor this week and next to discuss.

Week 7: Spillovers

Class 7.1

Estimation and design accounting for spillovers.

Readings

- Gerber, A. S., & Green, D. P. (2012). *Field experiments: Design, analysis, and interpretation*. WW Norton. Chapter 8: Interference between Experimental Units.

Class 7.2

Designs to detect spillovers.

Readings

- Miguel, E., & Kremer, M. (2004). Worms: identifying impacts on education and health in the presence of treatment externalities. *Econometrica*, 72(1), 159-217. https://doi.org/10.1111/j.1468-0262.2004.00481.x
  - Additional unrequired content on the "Worm Wars" regarding replication issues with Miguel & Kremer:
    * Vox article
    * Maintenance Phase podcast episode
- Wilke, A. M., Green, D. P., & Cooper, J. (2020). A placebo design to detect spillovers from an education–entertainment experiment in Uganda. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 183(3), 1075-1096. https://doi.org/10.1111/rssa.12571

Week 8: Instrumental variables

Class 8.1

Instrumental variables estimation.

Readings

- Angrist, J. D., & Krueger, A. B. (2001). Instrumental variables and the search for identification: From supply and demand to natural experiments. *Journal of Economic perspectives*, 15(4), 69-85. https://doi.org/10.1257/jep.15.4.69
- Athey, S., & Imbens, G. W. (2017). The econometrics of randomized experiments. In *Handbook of economic field experiments* (Vol. 1, pp. 73-140). North-Holland. https://doi.org/10.1016/bs.hefe.2016.10.003. Part 9.

Class 8.2

Applications with IV.

Readings

- De La O, A. L. (2013). Do conditional cash transfers affect electoral behavior? Evidence from a randomized experiment in Mexico. *American Journal of Political Science*, 57(1), 1-14. https://doi.org/10.1111/j.1540-5907.2012.00617.x
- Cameron, L., Erkal, N., Gangadharan, L., & Meng, X. (2013). Little emperors: behavioral impacts of China's One-Child Policy. *Science*, 339(6122), 953-957. https://doi.org/10.1126/science.1230221

Week 9: Heterogeneity and external validity

Class 9.1

Heterogeneous treatment effects and alternative experimental designs.

Readings

- Athey, S., & Imbens, G. W. (2017). The econometrics of randomized experiments. In *Handbook of economic field experiments* (Vol. 1, pp. 73-140). North-Holland. https://doi.org/10.1016/bs.hefe.2016.10.003. Parts 10-11.

Class 9.2

External validity.

Readings

- Banerjee, A., Duflo, E., Goldberg, N., Karlan, D., Osei, R., Parienté, W., ... & Udry, C. (2015). A multifaceted program causes lasting progress for the very poor:

Evidence from six countries. *Science*, 348(6236), 1260799. https://doi.org/10.1126/science.1260799

- Abaluck, J., Kwong, L. H., Styczynski, A., Haque, A., Kabir, M. A., Bates-Jefferys, E., ... & Mobarak, A. M. (2022). Impact of community masking on COVID-19: A cluster-randomized trial in Bangladesh. *Science*, *375*(6577), eabi9069. https://doi.org/10.1126/science.abi9069
- For reference: blog posts on meta-analyses
  - Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2019). Meaningless Means Series. *Data Colada*. Blog posts. https://datacolada.org/meaningless
- Not required: Meager, R. (2022). "Aggregating distributional treatment effects: A Bayesian hierarchical analysis of the microcredit literature." *American Economic Review* 112.6: 1818-1847. https://doi.org/10.1257/aer.20181811

Pre-analysis plans due on the last day of class.

Other readings

- Aronow, P. M., & Samii, C. (2017). Estimating average causal effects under general interference, with application to a social network experiment. *The Annals of Applied Statistics*, 11(4), 1912-1947. https://doi.org/10.1214/16-AOAS1005
- Athey, S., Bergstrom, K., Hadad, V., Jamison, J. C., Ozler, B., Parisotto, L., & Sama, J. D. (2021). Shared Decision-Making: Can Improved Counseling Increase Willingness to Pay for Modern Contraceptives?. Research Papers 3987, Stanford University, Graduate School of Business. https://doi.org/10.1596/1813-9450-9777
- Baird, S., Bohren, J. A., McIntosh, C., & Özler, B. (2018). Optimal design of experiments in the presence of interference. *Review of Economics and Statistics*, 100(5), 844-860. https://doi.org/10.1162/rest_a_00716
- Bowles, S., & Gintis, H. (2004). Homo Economicus and Zoon Politikon.
- Cameron, L., Meng, X., & Zhang, D. (2019). China's sex ratio and crime: Behavioural change or financial necessity?. *The Economic Journal*, 129(618), 790-820. https://doi.org/10.1111/ecoj.12561
- De Figueiredo, M. F., Hidalgo, F. D., & Kasahara, Y. (2011). When do voters punish corrupt politicians? Experimental evidence from Brazil. *Unpublished manuscript*, UC Berkeley.
- Fehr, E., & Fischbacher, U. (2002). Why social preferences matter–the impact of non-selfish motives on competition, cooperation and incentives. *The economic journal*, 112(478), C1-C33. https://doi.org/10.1111/1468-0297.00027

- Gonzalez-Ocantos, E., De Jonge, C. K., Meléndez, C., Osorio, J., & Nickerson, D. W. (2012). Vote buying and social desirability bias: Experimental evidence from Nicaragua. American *Journal of Political Science*, 56(1), 202-217.
- Ichino, N., & Schündeln, M. (2012). Deterring or displacing electoral irregularities? Spillover effects of observers in a randomized field experiment in Ghana. *The Journal of Politics*, 74(1), 292-307. https://doi.org/10.1017/S0022381611001368
- Mansuri, G., & Rao, V. (2013). Can participation be induced? Some evidence from developing countries. *Critical Review of International Social and Political Philosophy,* 16(2), 284-304. https://doi.org/10.1080/13698230.2012.757918
- Reinikka, R., & Svensson, J. (2005). Fighting corruption to improve schooling: Evidence from a newspaper campaign in Uganda. *Journal of the European economic association*, 3(2-3), 259-267. https://www.jstor.org/stable/40004969